



MANIPAL INSTITUTE OF TECHNOLOGY

MANIPAL

(A constituent unit of MAHE, Manipal)

COURSE PLAN

Department

:

MATHEMATICS

Course Name & code

:

ENGINEERING MATHEMATICS IV & MAT2204

Semester & branch

:

IV SEMESTER & CHEMICAL/ BIOTECH ENGG.

Name of the faculty

:

DR. SABITHA D'SOUZA

No of contact hours/week:

3 0 0 3

ASSESSMENT PLAN

Course Outcomes (COs)

At the end of this course, the student should be able to:

		No. of Contact Hours	Marks
CO1:	Solve boundary value problems of ODE and PDE using finite difference method.	9	12
CO2:	Understand the relevance of probability in engineering problems and explain the concepts of random variable.	8	10
CO3:	Identify situations of discrete/continuous probability distributions and apply these to approximate statistical parameters.	8	10
CO4:	Formulate the linear programming problems, solve by graphical, simplex, penalty cost or two phase methods.	9	10
CO5:	Solve difference equations by Z-transforms.	4	8
Total		38	50

Components	Surprise Quizzes	Sessional Tests	End Semester/ Make-up Examination
Duration	20 to 30 minutes	60 minutes	180 minutes
Weightage	20 % (4 X 5 marks)	30 % (2 X 15 Marks)	50 % (1 X 50 Marks)
Typology of Questions	Understanding/ Comprehension; Application; Analysis; Synthesis; Evaluation	Knowledge/ Recall; Understanding/ Comprehension; Application	Understanding/ Comprehension; Application; Analysis; Synthesis; Evaluation
Pattern	Answer one randomly selected question from the problem sheet (Students can refer their class notes)	MCQ: 10 questions (0.5 marks) Short Answers: 5 questions (2 marks)	Answer all 5 full questions of 10 marks each. Each question may have 2 to 3 parts of 3/4/5/6/7 marks
Schedule	4, 7, 10, and 13 th week of academic calendar	Calendared activity	Calendared activity
Topics Covered	Quiz 1 (L 1- 12 & T _{y1-y2}) (CO 1)	Test 1 (L 1 & T 17) (CO 1&2)	Comprehensive examination covering full syllabus. Students are expected to answer all questions (CO 1-5)
	Quiz 2 (L 13-20 & T _{y3-y4}) (CO 2 &3)		
	Quiz 3 (L 21-28 & T _{y5-y6}) (CO 3 &4)	Test 2 (L 18 & T 28) (CO 3&4)	
	Quiz 4 (L 29-38 & T _{y7-y8}) (CO 5)		

Course Plan

L. No./ T. No.	Topics	Course Outcome Addressed
L0	Introduction of the course. Numerical solution of ODE by finite difference method	CO1
L1	Numerical solution of ODE when boundary conditions involve derivatives.	CO1
L2	Numerical solution of PDE, Classification of PDE, Five point formula.	CO1
L3	Solution of Elliptic equation-Laplace and Poisson Equations, Problems.	CO1
L4	Tutorial	CO1
L5	Solution of Parabolic equations- heat equations using explicit and implicit methods	CO1
L6	Problems on heat equation using Crank-Nicolson method	CO1
L7	Solution of hyperbolic equations – wave equations	CO1
L8	Tutorial	CO1
L9	Finite sample spaces, Axioms of probability, Mutually exclusive events	CO2
L10	Conditional probability and independent events	CO2
L11	Tutorial	CO2

L12	Total probability theorem and Bayes' theorem	CO2
L13	One dimensional random variables, mean and variance	CO2
L14	Tutorial	CO2
L15	Chebyshev's inequality, uniform distribution.	CO2
L16	Two dimensional random variables.	CO2
L17	Tutorial	CO2
L18	Covariance, correlation coefficient	CO3
L19	Regression, least squares principle of curve fitting	CO3
L20	Tutorial	CO3
L21	Binomial and Poisson distributions	CO3
L22	Exponential, Gamma and normal distributions	CO3
L23	Problems on normal distribution and Chi-square distribution	CO3
L24	Tutorial	CO3
L25	Formulation of Linear Programming problem	CO4
L26	Graphical method of solving linear programming problem	CO4
L27	Tutorial	CO4
L28	Simplex method	CO4
L29	Problems based on Simplex method	CO4
L30	Tutorial	CO4
L31	Penalty cost method	CO4
L32	Two phase method	CO4
L33	Difference equations representing physical systems	CO5
L34	Properties of Z transforms, Initial and final value theorems	CO5
L35	Solution of difference equations by the method of Z transforms	CO5
L36	Problems on Z transforms, Convolution theorem	CO5
L37	Tutorial	CO5

References:

1. Sastry S.S - Introductory methods for Numerical Analysis. 2nd edn. Prentice Hall.
2. Meyer P.L. - Introduction to probability and Statistical applications, 2nd edn. American Publishing Co.
3. Hamdy A Taha - Operations research , 7th Edition, PHI.
4. Kreyszig E -.Advanced Engineering Mathematics, 7th edn. , Wiley Eastern.
5. Grewal B.S - Higher Engineering Mathematics, Khanna Publishers.
6. Richard Bronson- Operations Research- Schaum Series.
7. Narayanan, Ramaniah and Manicavachagam Pillay, Advanced Engineering Mathematics, Vol 3.

Submitted by: DR. SABITHA D'SOUZA

(Signature of the faculty)

Date: 01-02-2020

Approved by: DR.SUDHAKARA. G

(Signature of HOD)

Date: 02-02-2020

FACULTY MEMBERS TEACHING THE COURSE (IF MULTIPLE SECTIONS EXIST):

FACULTY	SECTION	FACULTY	SECTION
DR. RAVISHANKARA S BHAT	A		
DR. SABITHA D'SOUZA	B		
DR. HARINAKSHI KARKERA	BIOTECH		
